

# Benji Warburton

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UK citizen | F-1 student | CPT-eligible for full-time internships

## EDUCATION

### Stanford University

Stanford, CA

*B.S. Electrical Engineering and Computer Science; GPA: 3.7/4.0*

*Expected June 2028*

- **Coursework:** Reinforcement Learning\*, AI-Enabled Robotics, NLP with Deep Learning\*, Operating Systems, Digital System Design, Linear Algebra, Multivariable Calculus, Mechatronics, Signals and Systems, Advanced Mechanics with Special Relativity, Computer Systems, Circuits. \**Graduate-level course.*

## EXPERIENCE

### Apple

Jun. 2026 – Sep. 2026

*Hardware Technology - Engineering Intern*

*Sunnyvale, CA*

- Developing a **GNSS simulation platform** with senior engineers; generating MATLAB and C/C++ modules and test vectors for unit-level verification of signal-processing components.
- Supporting end-to-end GNSS firmware validation and documenting system design and experimental results.

### Y Combinator

Jun. 2026 – Sep. 2026

*Summer Fellow - Low-Cost Robot Actuators*

*San Francisco, CA*

- Designed and fabricated robot-joint actuators using frameless motors, off-the-shelf ESCs, and 3D-printed gearboxes, achieving **approximately 6× lower cost** with comparable performance.
- Built a programmable stator-winding machine and designed a custom ESC while moving motor, gearbox, electronics, and test infrastructure toward in-house fabrication.

### Stanford Assistive Robotics and Manipulation Lab (ARMLab)

Feb. 2026 – Present

*Research Assistant, advised by Prof. Monroe Kennedy III*

*Stanford, CA*

- Developing **TidyBot**, an embodied-AI system that scans rooms, recognizes individual objects, and restores displaced items to their original poses using coordinated manipulation and locomotion policies.

### Stanford CS123: Introduction to AI-Enabled Robotics

Sep. 2025 – Present

*Teaching Assistant and Curriculum Developer*

*Stanford, CA*

- Only undergraduate on an MS/PhD/postdoc teaching team led by Prof. Karen Liu; created labs with collaborators from Google DeepMind and mentor 6 teams across control, perception, RL, and sim-to-real.

### Renaissance Philanthropy

Aug. 2025 – Oct. 2025

*Student Fellow*

*Remote*

- Built a parallel, cached Python ETL and ranking pipeline over **80M+ works and 400K+ authors**, producing field-normalized impact metrics and QC dashboards to surface high-potential researchers.

### YASA Motors

Jul. 2023

*Data Science Intern*

*Oxford, UK*

- Analyzed electric-motor dyno end-of-line tests to shorten test cycles and helped identify a measurement issue inflating cogging-torque values.

## SELECTED RESEARCH & PROJECTS

### From Text to Torque: Language-to-Humanoid Policies | *diffusion, PPO, MuJoCo Warp, Unitree G1*

2026

- Built an end-to-end pipeline translating natural-language commands into **diffusion-generated motions**, then training PPO control policies runnable on a simulated Unitree G1 humanoid.
- Scaled MuJoCo Warp training across **1,024 parallel environments**; designed adaptive termination calibration that recovered full **250-step** squat tracking where default strict training averaged 13.5 steps.

### 5-DOF Robot Arm | *ROS 2, C++, NVIDIA Isaac Lab, GR00T N1, custom hardware*

2025 – 2026

- Ran NVIDIA's **2B-parameter GR00T N1 vision-language-action (VLA) model** on an RTX 3080 and connected inference to a Raspberry Pi running the arm's ROS 2 control stack.
- Designed and 3D-printed the arm and wrote a C++ **ros2\_control** hardware interface for its TTL-bus servos.

### Compositional Tool-Sequence Generation in SLMs | *Qwen3, PyTorch, SFT*

2026

- Fine-tuned Qwen3 small language models for structured, multi-step **tool calling**; co-designed an instruction-to-program benchmark where SFT improved exact match by over **3×** and length accuracy by over **5×**.

### 8-Cell Active Li-Ion Battery Balancer | *power electronics, Arduino, Python*

2023

- Designed and prototyped active balancing hardware with pack logging and calibration; won the UK national **TDI Challenge** from the Manufacturing Technologies Association.

## TECHNICAL SKILLS & RECOGNITION

**Software:** Python, C/C++, MATLAB, PyTorch, ROS 2, Isaac Lab/Sim, MuJoCo, CUDA, Git, Linux, Verilog

**Hardware:** PCB and ESC design, CAN, embedded systems, motor testing, CAD, 3D printing, rapid prototyping

**Recognition:** RISE Global Winner (1 of 100; ~0.2% selection) for a sub-\$45 robotics education kit